

DWI denoising with DCT and PCA

Yiming(Eric) Cao
92138098

Li Guo
75279364

Yingrui Yang
79902243

Shengyu Zuo
23571953

Haoxiang Lei
27658459

Abstract—Diffusion-weighted imaging (DWI) is an advanced method of contrast generation based on the Brownian motion of water molecules (diffusion) within tissue voxels. Although DWI is powerful, diffusion-weighted images suffer a lot from their intrinsically poor signal-to-noise ratio (SNR). In this project, we will use two methods to denoise the DWI, first one is the Discrete Cosine Transform (DCT), which could remove the noise in the frequency domain. The second is the Principle Component Analysis (PCA), which denoises the image in the pixel domain. We perform these two methods on a dataset containing DWI images from 3 different subjects. The DCT denoising achieves an increase in SNR around 1.7 dB and the PCA achieves an increase around 5.1 dB. We discuss the results and conclude that PCA has an overall better performance over DCT in this specific application. We also indicate possible improvements on this project for future work.

Index Terms—MRI, DWI, denoise, PCA, DCT

I. INTRODUCTION

A. Background

The human body is made of water for around 60–70% of its total mass. The random Brownian motion of the molecules, which is driven by thermal energy, is what we refer to as diffusion. In a material that is entirely homogeneous, diffusion will be completely random and isotropic. This diffusion pattern is disrupted in various pathological circumstances, such as acute stroke, and the quantity of diffusion in the afflicted region varies. By carefully observing how the diffusion varies over time, it is possible to identify any irregularities. This may be accomplished using a particular kind of magnetic resonance imaging known as diffusion-weighted MRI (DW-MRI) or DWI [1].

Diffusion-weighted imaging (DWI) is an advanced method of contrast generation based on the Brownian motion of water molecules (diffusion) within tissue voxels. DWI pictures have the benefit of being able to clearly highlight the region that is affected by cerebral ischemia as well as tumors. It is possible to quantify DWI signal contrast using apparent diffusion coefficient maps. These maps serve as a useful tool for evaluating therapy responses and determining the development of illness.

B. Motivation and Problems

Although DWI is powerful, it has disadvantages. Diffusion-weighted images have an intrinsically poor signal-to-noise ratio (SNR) owing to low signal amplitude and significant thermal noise, which is more visible than in traditional MRI due to highly quick echo-planar recording procedures [2].

DWI analysis becomes more difficult with such a low SNR, and the calculation of quantitative diffusion parameters is

biased as a result. In addition, because of the low SNR, the automatic processing of these pictures is difficult and might result in inaccurate interpretations.

In the project, we mainly use two methods to denoise the DWI, first one is the Discrete Cosine Transform, which could remove the noise in the frequency domain. The second is PCA, which denoises the image in the pixel domain, we will expand on that later.

C. Related Works

The state-of-the-art denoising method is based on neural networks and deep learning. Cheng et al. introduce a one-dimensional convolutional neural network (1D-CNN) model and deep learning methods for denoising diffusion-weighted images [3]. The model is trained, voxel to voxel, using low-noise datasets and can be applied to high-noise images from other subjects. Their method focuses on removing the rectified noise floor from images with a very low SNR that most other denoising methods lack. They validate the performance of their model by using the in vivo dataset in DWI image, diffusion matrices, and tractography, where promising empirical results are generated. Thus, CNN denoising method is reckoned as a simple and effective method of DWI denoising.

Other methods based on optimization and statistical algorithms are also insightful and useful. Ding et al. apply an anisotropic smoothing method to denoise diffusion tensor images (DTI) [4]. By applying isotropic smoothing within homogeneous regions and anisotropic smoothing along structure boundaries of the DWI data, their method can therefore reduce the noise of the consequent derived DTI data. Experiments on simulated and in vivo diffusion tensor data are performed to quantitatively analyze the effectiveness of their method. Moreover, their method is considered to be more practical in DTI in which the direction information needs to be preserved around tissue boundaries. Cordero et al. propose a patch-based singular value shrinkage (SVS) method for DWI estimation targeted at low SNR and accelerated acquisitions [5]. They formulate the noisy DWI data as a set of matrix recovery problems via singular value decomposition (SVD) and shrinkage. Their model operates on the complex data resulting from sensitivity encoding reconstruction (SENSE) where noise propagation is obtained to consequently simulate the singular value spectrum.

The PCA method of our project is inspired by Zhang et al [6], where the noise and unimportant information is removed by transforming the datasets into the PCA domain with only the most significant principal components. Zhang

et al. proposed local pixel grouping PCA that a pixel and its nearest neighbors are modeled as a vector variable and the training samples are selected by grouping similar local structures. This procedure can preserve the information in the PCA domain after noise removal by computing the local statistics.

II. DATA SET

The dataset of DWI images is taken from the multishell diffusion MRI dataset collected from three traveling subjects with identical acquisition settings in ten imaging centers [7]. The imaging parameters were as follows: TR/TE = 5.4s/71ms, FOV = $220 \times 220 \text{ mm}^2$, slice number = 93, voxel size = $1.5 \times 1.5 \times 1.5 \text{ mm}^3$. The diffusion scheme contains 30 vectors with uniform angular coverage on each shell (b-values = 1000, 2000, and 3000s/mm², non-collinear between any two shells). Six non-diffusion frames were equally distributed in the scheme.

III. TECHNIQUES

To denoise an entire brain image, we need to intercept the cross-section of the 3D brain image slice by slice to a set of 2D images, and process the denoise method on each 2D image.

A. Image decompose to Patches

Before applying different denoising methods, we need to windowize each image, which is decomposing the whole image into multiple square areas. Each square area is a patch, and the order of patch selection is based on the order from left to right and then from top to bottom with a stride of one pixel.

Assuming that the side length of the patch is p , and there are a total of N patches in an image, which is the total pixels in an image. After the above preprocessing, we obtain a three-dimensional data of size (N, p, p) as the input of the denoise method.

Algorithm 1 Image to Patches

Input: Image I of size (a, b)
Output: Patches P of size (N, p, p)
 $pixel_pointer = (0, 0)$
 $P = []$
for Each row in I **do**
 for Each column in I **do**
 if $pixel_pointer.y == b - p$ **then**
 $pixel_pointer = pixel_pointer + (1, 0)$
 else
 $pixel_pointer = pixel_pointer + (0, 1)$
 end if
 select a matrix of size (p, p) and append to P
 end for
end for

B. Patches reconstruct to Image

After denoising, the size of the above 3D data remains unchanged, and we need to reconstruct this 3D data to a 2D image according to the above order. It is worth mentioning that the selection of each patch is to move the previous patch one pixel to the right, so there will be overlaps between adjacent patches. In the reconstruction step, we need to add weight to the overlapping pixels.

Algorithm 2 Patches to Image

Input: Patches P of size (N, p, p)
Output: Image I of size (a, b)
 $I = []$
 $I_weight = []$
for $i = 1$ to N **do**
 add patch $P(i)$ to I pixel by pixel if there is no overlappings
 if $pixel(m, n)$ is already in I **then**
 $I(m, n) = I(m, n) + pixel(m, n)$
 $I_weight(m, n) = I_weight(m, n) + 1$
 end if
end for
 $I = I / I_weight$

C. DCT

The image discrete cosine transform (DCT) transfers the pixels of an image to a sum of sinusoids with different magnitudes and varying frequencies. The output of a DCT module is an image in frequency domain with the value of each pixel representing the magnitude or the DCT coefficient, and the position of each pixel representing the frequency. Usually, the top-left corner represents low-frequency components and the bottom-right corner represents high-frequency components.

The two-dimensional DCT of an M-by-N image (matrix) A is defined as follow:

$$B_{pq} = \alpha_p \alpha_q \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} A_{mn} \cos \frac{\pi(2m+1)p}{2M} \cos \frac{\pi(2n+1)q}{2N}$$

$$\text{where, } \begin{cases} 0 \leq p \leq M-1 \\ 0 \leq q \leq N-1 \end{cases}$$

$$\alpha_p = \begin{cases} 1/\sqrt{M}, & p = 0 \\ \sqrt{2/M}, & 1 \leq p \leq M-1 \end{cases}$$

$$\alpha_q = \begin{cases} 1/\sqrt{N}, & q = 0 \\ \sqrt{2/N}, & 1 \leq q \leq N-1 \end{cases}$$

By traversing through each pixel of the output image B (varying the value of p and q , B_{pq} , the DCT coefficients can be generated. The α_p and α_q are normalizing factors.

Also, the DCT is an invertible algorithm. The inverse DCT (IDCT) can be performed on the frequency domain image to transfer it back to the pixel domain. The inverse algorithm is given by the following equations:

$$A_{mn} = \sum_{p=0}^{M-1} \sum_{q=0}^{N-1} \alpha_p \alpha_q B_{pq} \cos \frac{\pi(2m+1)p}{2M} \cos \frac{\pi(2n+1)q}{2N}$$

$$\text{where, } \begin{cases} 0 \leq m \leq M-1 \\ 0 \leq n \leq N-1 \end{cases}$$

$$\alpha_p = \begin{cases} 1/\sqrt{M}, & p = 0 \\ \sqrt{2/M}, & 1 \leq p \leq M-1 \end{cases}$$

$$\alpha_q = \begin{cases} 1/\sqrt{N}, & q = 0 \\ \sqrt{2/N}, & 1 \leq q \leq N-1 \end{cases}$$

In this project, the DCT is performed on each patch, the output of the preprocessing module with a size of 8 by 8, that is, $M = 8, N = 8$ in the above equations. After B along with the coefficients B_{pq} are obtained from DCT, the high-frequency components, that is, the coefficients in the bottom-right corner of B , are set to 0 to achieve denoising. More specifically, only the top-left d by $d+1$ coefficients are preserved, and all the other coefficients are set to 0, where d can be ranged from 1 to 7. A more detailed explanation of choosing the value of d will be discussed in the result section. This is done because noise is normally in high-frequency components and the main power of the image is in low-frequency components, therefore, setting high-frequency components to zeros achieves denoising while preserving the main content of the image. This process is essentially a low-pass filter performed in the frequency domain. After denoising, the new B matrix is fed through the IDCT algorithm to retrieve a denoised patch in the pixel domain. By repeating the above DCT-denoising-IDCT process for every input patch, a group of denoised patches can be retrieved. Finally, these denoised patches are regrouped to form a result image and this image should be the denoised version of the original one.

For this project, a DCT denoising algorithm is performed for patches with size ($p=8, p=8$) from the 3D data with size (N, p, p) in the preprocessing method mentioned earlier, this algorithm is shown in Algorithm 3.

Algorithm 3 DCT

Input: Patches P of size $(N, p, p), k_1, k_2$

Output: Denoised patches P' of size (N, p, p)

```

for  $i = 1$  to  $N$  do
    DCT transformation on  $P(i)$ 
    keep  $k_1$  row and keep  $k_2$  column  $P(i)$  to remove high
    frequency component
    inverse DCT transformation on  $P(i)$ 
end for

```

D. PCA

Principal component analysis (PCA) is a process of dimensionality reduction and decorrelation. In this process, PCA first removes components that are less correlated with other components. Pixels in the same tissue have similar intensities, and the intensity of noise is less correlated with the intensity of others, so noise components are preferentially removed. The idea of denoising with PCA was inspired by LPG-PCA, a noise reduction model proposed by Zhang et al.

In our PCA denosing method, we converted each patch

(p, p) of the 3D data $(N, p \times p)$ into a vector $(1, p \times p)$ and then superimposed with the original order to form a new data X with size $N \times p^2$. After that, performing PCA transformation on X .

Then, applied singular value decomposition (SVD) on the matrix X ,

$$X = U \Sigma V^T$$

where U was a $N \times N$ complex unitary matrix, Σ was a $N \times p^2$ rectangular diagonal matrix with non-negative real numbers on the diagonal, V was a $p^2 \times p^2$ complex unitary matrix, and V^T was the transpose of V .

Let $Z = U \Sigma$ and $W = V^T$ as shown in Figure 1.

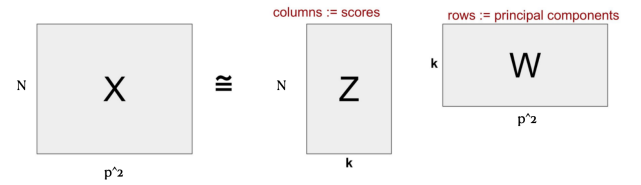


Fig. 1. Process of PCA.

where k was the number of principal components to keep and it ranged from 1 to p^2 . By reducing the value of k , the information of the original image could be discarded. The choice of k value should not only retain enough structural information of the original image, but also remove the noise information. The recommended value range of k was shown in the results. After choosing an appropriate value of k , multiplied Z and W to obtain $\tilde{X}$.

For this project, a PCA denoising algorithm is performed for patches with size ($p=5, p=5$) from the 3D data with size (N, p, p) in the preprocessing method mentioned earlier, this algorithm is shown in Algorithm 4.

Algorithm 4 PCA

Input: Patches P of size $(N, p, p), k$

Output: Denoised patches P' of size (N, p, p)

```

 $X = []$ 
for  $i = 1$  to  $N$  do
    resize  $P(i)$  to  $(1, p * p)$  and append to  $X$ 
end for
for Each column vector  $X_i \in X$  do
     $X_i = X_i - \mu_{x_i}$ 
end for
 $SVD(X) = U \times S \times V^T$ 
 $W = V^T[:, k]$ 
 $Z = X \times W^T$ 
 $X' = Z \times W$ 
 $P' = []$ 
for  $i = 1$  to  $N$  do
    resize  $X'(i)$  to  $(p, p)$  and append to  $P'$ 
end for

```

IV. METRICS

A. SNR

The SNR was measured following the National Electrical Manufacturers Association (NEMA) standard on two adjacent slices. Because the distance between two adjacent slices was small enough, we assumed that the images of these slices were acquired at the same time and under the same conditions, which means that the two images were identical. In order not to make the blurry image have a significantly high SNR, we used the whole brain as the region of interest (ROI) instead of only one white matter region at the genu of the corpus callosum that was usually used. In favor of obtaining a sufficiently large ROI, for each DWI image, we only used the middlemost two slices to calculate the SNR. The mean values of two images within ROI were termed the signal. The noise was calculated from the standard deviation (SD) of the difference image within the same region divided by a correction factor of $\sqrt{2}$.

B. SSIM

The Structural Similarity Index (SSIM) is a perceptual metric that quantifies image quality degradation caused by processing. In this project, SSIM was used to measure the similarity of images before and after denoising. Small SSIM meant that the denoised image was blurred and lose the structural information of the original image.

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)}$$

with:

- x the original image;
- y the denoised image;
- μ_x the pixel mean of x ;
- μ_y the pixel mean of y ;
- σ_x^2 the variance of x ;
- σ_y^2 the variance of y ;
- σ_{xy}^2 the covariance of x and y ;
- $c_1 = (k_1L)^2$, $c_2 = (k_2L)^2$ two variables to stabilize the division with weak denominator;
- L the dynamic range of the pixel-values (typically this is $2^{\#bits \text{ per pixel}} - 1$);
- $k_1 = 0.01$ and $k_2 = 0.03$ by default.

V. RESULTS

To evaluate the noise reduction of DCT and PCA, and select appropriate k and d . The following two line charts in Fig. 2 and 3 show the trend of two metrics, SNR and SSIM of the two methods in the range of all k and d respectively.

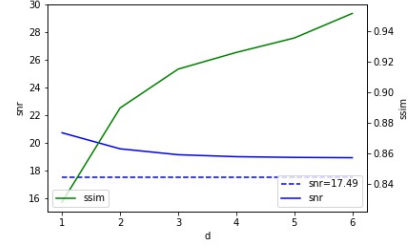


Fig. 2. DCT

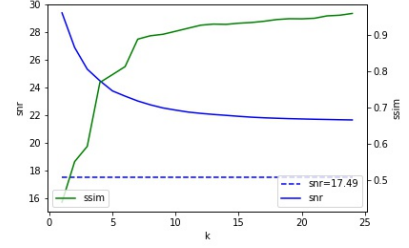


Fig. 3. PCA

As shown in Fig. 2 and 3, The dashed line is the SNR of the image before denoising, the blue line represents the SNR of the image after denoising, and the green line is the SSIM of the images before and after denoising. As the values of k and d increases, the corresponding SNR decreases and SSIM improves.

A low SSIM is a signal that the denoised image is blurred, and a high SSIM implies that the denoised image retains redundant information of the pre-denoised image including noise. Therefore, the goal is to pursue high SNR while keeping SSIM in a suitable range. After experiments, through human perception scoring, when the SSIM was between 0.88 and 0.93, the denoised image not only retained enough structural information, but also had an obvious denoising effect. Therefore, it is necessary to restrict the value of k and d to keep SSIM within the desired range. For DCT method, when the image patch is 8 by 8, the recommended range of d is [2, 5]. For PCA method, when the image patch is 5 by 5, the recommended range of k is [8, 16].

Table I, II and III respectively shows the experimental results of 3 subjects in the dataset mentioned earlier at different b-values. The 'Difference' columns after the DCT and PCA mean the difference between the SNR of the image before denoising and the image after DCT processing and PCA processing, respectively. It can be seen from the three tables that both DCT and PCA methods can stably denoise DWI images. The larger the b value, the more the SNR of the denoised image is improved. Therefore, these two methods have better noise reduction effects for DWI images with larger b values.

In addition, for DWI images from different people in different MRI machines, PCA method can increase SNR by

TABLE I
COMPARISON TABLE FOR SUB-1

SNR(dB)	Original	DCT	Difference	PCA	Difference
b=0	14.4±0.2	15.8±0.3	1.4±0.1	18.8±0.3	4.4±0.1
b=1000	16.8±0.2	18.4±0.2	1.6±0.1	21.5±0.1	4.7±0.1
b=2000	16.0±0.3	17.7±0.3	1.7±0.1	21.1±0.5	5.1±0.2
b=3000	16.3±0.1	18.1±0.1	1.8±0.1	21.6±0.1	5.2±0.1

TABLE II
COMPARISON TABLE FOR SUB-2

SNR(dB)	Original	DCT	Difference	PCA	Difference
b=0	15.7±0.4	17.3±0.4	1.6±0.1	20.9±0.4	5.2±0.1
b=1000	17.1±0.4	18.9±0.4	1.7±0.1	22.8±0.4	5.7±0.1
b=2000	16.7±0.1	18.9±0.3	2.1±0.3	22.7±0.1	5.9±0.2
b=3000	16.5±0.2	18.5±0.1	1.9±0.1	22.7±0.3	6.2±0.3

TABLE III
COMPARISON TABLE FOR SUB-3

SNR(dB)	Original	DCT	Difference	PCA	Difference
b=0	16.3±0.2	17.8±0.2	1.5±0.1	21.2±0.1	5.0±0.1
b=1000	17.7±0.3	19.4±0.2	1.6±0.1	22.9±0.1	5.2±0.2
b=2000	17.1±0.4	18.8±0.4	1.7±0.1	22.5±0.2	5.4±0.2
b=3000	16.6±0.2	18.8±0.2	2.1±0.3	22.7±0.1	5.7±0.2

4.7 to 5.7 dB and DCT method can increase SNR by 1.6 to 2.1 dB. Thus, for DWI images, the PCA method has significantly better noise reduction ability compared with the DCT method. It can be clearly seen in Figure 4, where the PCA-processed image retains fewer black spots.

is a chance to do further research on this project, we hope to be able to use deep neural networks to directly denoise the whole 3D image instead of 2D slice, which may preserve more spatial information of the image.

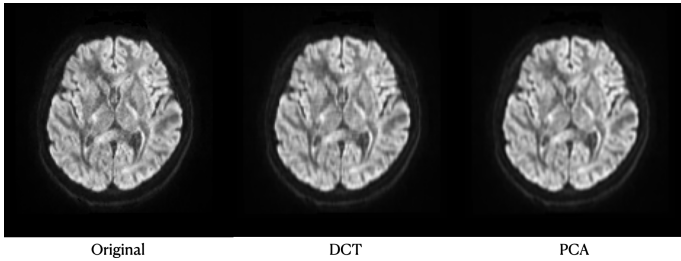


Fig. 4. Denoised images of DCT and PCA

VI. CONCLUSION

In general, our project is successful. We use two methods to improve the SNR by around 1-5 dB for noise reduction. the DCT method improves SNR by around 1-2 dB, and the PCA method improves SNR by around 4-5 dB. Therefore, we conclude that PCA is better than DCT in this application.

In the experiment, two problems are found. First, whether using DCT or PCA, the overall quality of the image decreases while denoising, and the better the noise reduction, the lower the image quality. Second, our denoising method is used on 2D images, and because of the first problem, when reconstructing 2D images into 3D images, some image components will lose. Our future work is to optimize these two problems. If there

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APPENDIX

As biomedical electrical engineers, we thrive for making innovative and practical contributions to today's technological development. Our project proposes a meaningful insight into improving the imaging performance of MRI based on preceding studies, which could hopefully inspire other people in relative future studies. As we work together, we build effective and harmonic teamwork where the assignments are equally distributed with every members' consent. We take full responsibility for the outcome and risk of our idea to anyone's unsatisfactory or disagreement and we acknowledge the possible deficiency of our point of view. We show great respect for academic integrity and we guarantee our work is completely genuine.